Assignment 10

Chapter 8

4. *Product source* Let X and Y be two independent source random variables with reproduction alphabets $\hat{\mathcal{X}}$ and $\hat{\mathcal{Y}}$ and distortion measures d_x and d_y, and the rate-distortion functions for X and Y are denoted by $R_x(D_x)$ and $R_y(D_y)$, respectively. Now for the product source (X, Y), define a distortion measure $d : \mathcal{X} \times \mathcal{Y} \to \hat{\mathcal{X}} \times \hat{\mathcal{Y}}$ by

$$d((x, y), (\hat{x}, \hat{y})) = d_x(x, \hat{x}) + d_y(y, \hat{y}).$$

Prove that the rate-distortion function $R(D)$ for (X, Y) with distortion measure d is given by

$$R(D) = \min_{D_x + D_y = D} (R_x(D_x) + R_y(D_y)).$$

Hint: Prove that $I(X, Y; \hat{X}, \hat{Y}) \geq I(X; \hat{X}) + I(Y; \hat{Y})$ if X and Y are independent. (Shannon [327].)

6. Show that asymptotic optimality can always be achieved by separating rate-distortion coding and channel coding when the information source is i.i.d. (with a single-letter distortion measure) and the channel is memoryless.

Chapter 9

3. Explain why in the BA Algorithm for computing channel capacity, we should not choose an initial input distribution which contains zero probability masses.